

1. WHO? KNOW YOUR AUDIENCE

- a. List the primary group of individuals to whom you'll be communicating.

Marketing Department Leaders.

- b. If you had to narrow down that to a single person, who would that be?

Chief Marketing Officer

- c. What does your audience care about?

Our audience needs the attributes and indicators based on which the tele-marketing team must prioritize reaching out to potential customers such that it enables to **maximize their efficiency** by allowing them to spend maximum efforts where the **probability of subscription** of term deposit by the potential customers is higher than the present process followed.

- d. What action does your audience need to take?

The bank marketing team needs to filter the available data by job type and select the attribute as: **Student and Retired**.

In customer with **Job=Student**, the marketing team needs to **filter by age** and fetch the records **up to 30 years** and prioritize those customers.

In customers with **Job=Retired**, the marketing team needs to filter by age and fetch the records **greater than 60 years** and prioritize those customers.

As the result, we want the bank's marketing team to initially focus on customers which fall under two categories of Job: Student and Retired, subclassified with respect to their Age. And maximize the potential of this particular segment of customers by investing maximum focus and efforts over them and leveraging the potential of the specific customer segment.

Lastly, the marketing team should prioritize to reach out the potential customers on Cellular over Telephone.

- e. What is at stake? What is the benefit if the audience acts in the way you want them to? What are the risks if they don't?

What is at stake?

- For a bank, as more customers get onboarded and invest their money in term deposit, the bank maintains a constant influx of capital which allows them to utilize it for earning more, managing the expenses and allow to give returns on investment to its customers.
- Thus, the bank will look towards: -
 - Improving the operational efficiency of marketing team
 - **Reduction of cost incurred in marketing** while getting better results.

What is the benefit if the audience acts in the way you want them to?

- As per the given dataset, out of the total number of potential customers 41188, 4640 subscribed for term deposit.
- Thus, on average 11.26% of the total customers subscribed for term deposit.
- If the bank understands the attributes of those 11.26% potential customers and accordingly alter their list of customers to reachout via tele-marketing, then the bank will have better probability of onboarding more customers in lesser time.
- Thus, the overall marketing efficiency improves, costs incurred reduce and bank gets to target the specific customer segments and strengthen that area to achieve maximum potential.
- If marketing team focuses on the subset suggested, they will have around 40% successful onboarding of customers out of the total they reach-out during the campaign. In absolute numbers, as per the given dataset, the **revised count of total number of customers** to reachout [job=student and age <= 30 years + job=retired and age > 60 years] are **1489**. Out of 1489, **598 customers subscribed** for the term deposit.
- Thus, we draw a simple comparison between the present approach and the revised approach.
 - i) In the present approach, if marketing team reaches out to 100 potential customers, they will get 11 customers who subscribe for the term deposit.
 - ii) In the revised approach, if marketing team reaches out to 100 potential customers, they will get 40 customers who subscribe for the term deposit.Thus, we find higher probability of success in the revised approach.
- Since our count of target customers will be lesser than the original dataset, the tele-marketing team will be able to get more customers in lesser number of calls, and the time saved can be used to survey and get details of new customers who fall in those categories.
- As the result, the bank's marketing team will be able to focus and find a specific subset of potential customers and carry out their tele-marketing campaign keeping those customers at highest priority to ensure maximum number of those customers are reached out.

What are the risks if they don't?

- As per the current process, the marketing team randomly selects customers from 41188 and reaches out to them.
- The probability of reaching out to the customer who actually subscribes for the term deposit is relatively very lower than the probability of reaching out to the customers who actually do not subscribe for the term deposit.
- Thus, the marketing team spends more time on those customers who will not subscribe for the service than those customers who will subscribe for the service.
- As the result, the customers acquisition will get delayed and other competitive banks may take away the customers who could have been onboarded by our bank. Thus, there is loss in terms of converting potential customers to new customers of the bank.

2. WHAT?

- What are you trying to communicate?

Q1 Who do we primarily target during the marketing campaign?

We want to primarily reach out customers whose job is: student and retired.

Q2 Which age group do we target?

For students, we should reach out whose age is less than 30 years.

For retired, we should reach out whose age is greater than 60 years.

Q3 How should we reach out during the marketing campaign?

We should prioritize to reach out all customers via cellular over telephone.

- Data preparation needed to answer the specific queries must be done.

3. Present the BIG IDEA?

Optimizing tele-marketing efforts by focusing on students and retired customers while prioritizing cellular outreach, to enhance the conversion rate of phone conversations into purchase of the product.

4. HOW?

- a. Chart 1: What type of visualization did you create? Why did you select the visualization?

- We created a **Text table visualization** to display percentage of customers who opted for Term Deposit, categorized by their Job.
- The table is sorted in decreasing order of percentage values, thus, we observed among **total student customers** in the given dataset, **31.43% opted** for Term Deposit.
- Similarly, among the **total retired customers** in the given dataset, **25.23% opted** for Term Deposit.
- **Reason for using the visualization: -**
 - i) It gives a simple and summarized view of available dataset.
 - ii) The job categories of interest are highlighted in green to get faster attention.
 - iii) The top down descending order helps to easily identify and distinguish the job categories and their contribution towards term deposit subscription. It prevents the reader to scroll mouse or eyes up and down to comprehend the values and reader can read in one direction.

- b. Chart 2: What type of visualization did you create? Why did you select the visualization?

- We created a **Heat map visualization** to display percentage of customers who opted for Term Deposit, categorized by job: retired and student and then sub-categorized by their age range.

- The table has age range of 10 years; thus, it allows to get a brief overview of distribution of Term Deposit subscription over different age groups among students and retired customers.
- From the table we can derive that out of every 10 students subscribing for term deposit, 7 are in the age group of 21 to 30 years.
- From the table we can comprehend among the retired customers subscribing for term deposit, most of them have age above 50 years.
- **Reasons for using the visualization: -**
 - i) Heat map easily helps to visualize and distinguish the upper percentage values and lower percentage values.
 - ii) Using bins for Age helps to group the data into similar categories and thus, the visualization becomes more easily readable without the requirement of horizontal and vertical scrolling.
 - iii) The use heat map visualization allows to immediately identify the data points of interest from the table based on strength of the shade of colour.

c. Chart 3: What type of visualization did you create? Why did you select the visualization?

- We created a **Stacked bar chart visualization** to display percentage of customers who opted for Term Deposit, categorized by marital status and sub categorized by method of contact employed: cellular and telephone.
- From the chart we can derive that irrespective of the customer's marital status, contact using cellular was having better results than telephone.
- **Reasons for using the visualization: -**
 - i) The visualization helps us infer that for customers irrespective of their marital status, proportion of cellular is higher than telephone.
 - ii) The visual chart helps us interpret the data with two colours, thus, using minimum number of colours for effective visualization.
 - iii) It is a categorical vs categorical visualization which is effectively represented using stacked bar chart.

d. For each of the visualization, identify at least 3 **Gestalt principles** employed.

- **Chart 1: -**
 1. Law of Pragnanz (Simplicity)
 2. Law of Focal Point
 3. Law of Proximity
- **Chart 2: -**
 1. Law of Pragnanz (Simplicity)
 2. Law of Similarity
 3. Law of Continuity
- **Chart 3: -**
 1. Law of Pragnanz (Simplicity)
 2. Law of Similarity
 3. Law of Proximity

- e. For each of the visualization, mention how you strategically used **pre-attentive attributes** to draw the audience's attention.

- **Chart 1: -**

1. We used **green colour** to prioritize the attention of the user on specific datapoints in the table.
2. It will create **pre-attentive signals** to quickly draw user's attention where we need them to look at the first instance.
3. We used the concept of **Information hierarchy** by ordering the dataset based on percentage of values in decreasing order.
4. It is the key driver of our dashboard's insights and thus, it is placed in top left section of the dashboard so that reader first views the key information and then goes further to other supporting details.

- **Chart 2: -**

1. We used **heatmap** which brings different shades of blue colour based on **magnitude of values** in the table to prioritize the attention of the user on most important datapoints in the table.
2. By using different shades of the same colour, we **build pre-attentive signals** by creating **visual hierarchy** of the information and thus, allowing the user to immediately relate with the datapoints in the table prior reading the actual values in it.
3. We used the concept of Information hierarchy by **ordering the dataset based on age range** from top to bottom in increasing order.

- **Chart 3: -**

1. We have used two specific **colours to distinguish between contact type** in each stacked bar. Thus, it helps to easily relate with the interpretation of the graph.
2. The **bar chart's length** helps to **easily compare** and **identify the trend** in the dataset.

Assumptions and limitations of the analysis: -

1. Imbalanced nature of the dataset: - We have not changed the nature of dataset which is originally highly imbalanced and explicitly gives discriminating relation between two classes of the target feature.
 - The dataset is highly imbalanced with respect to the subscription of term deposit by customers.
 - This may contribute towards errors in the accuracy of analysis of the dataset and missing out of important patterns and insights during the analysis.
 - There are many techniques used for handling imbalanced nature such as SMOTE, Oversampling.
 - SMOTE will require use of external tools and libraries and cannot be carried out in Tableau.
 - Oversampling contributes towards changing the size of existing dataset and due to appending of duplicate records we may bias the analysis by using the modified dataset. Additionally, it will lead to diversion of behaviour of the collected sample-customer dataset from behaviour of the overall population of customers in the real-world.

2. Limited use of features: - As per findings documented above, we have restricted the selection of customers based on their job and age because it enables us get relatively higher probability of successful subscription as compared to current probability.
 - Out of 20 independent features, our decisions to identify potential customers are restricted based on two features, thus, our analysis may miss out many other possible important information and trends in the dataset.
 - We carried out **univariate analysis** of each categorical feature and **bi-variate analysis** of each categorical feature with the target. We assume due to **imbalanced nature of the dataset**, all categorical features **displayed similar pattern** where the **undesired class of target is relatively very large** in count.
 - We also transformed numerical features into ranges and plotted them individually with target, and also tried to analyze the same with other categorical features, but we did not find any insights which could help us determine the factors that can be used for making the final decision about selection of prospective customers.
3. Target variable is chosen as subscription of Term Deposit: - As per the problem statement, we need to analyze the dataset for finding the factors which contribute to successfully improve conversion of telemarketing into purchase of products offered by the bank.
 - A bank can offer various products as part of its diversified product portfolio: Loan, Housing Loan, Term Deposit.
 - However, as per the additional document shared with title “bank-additional-names.txt”, we have assumed Term Deposit as our target feature, and rest of the features as independent.
4. Usage of field “Duration”: - As per the additional document given along with the assignment, this attribute highly affects the output target should be included for benchmark purposes. However, in our analysis, we have not considered it because: -
 - We will know the outcome of field: duration only after the call has been made to the customer.
 - Thus, this is not seen as an attribute of the customer prior reaching out to them.
5. Usage of social and economic context attributes: - Five attributes related to social-economic attributes were listed but they required deeper domain level understanding of relation with respect to Term Deposit, thus, we do not consider them in our current analysis.
6. Imputation of missing values in categorical fields.
 - If we check all categorical features and filter out value: **unknown**, then we are **left with 38311 out 41188 records**.
 - Thus, we will end up losing 2877 records of the given list of customers which accounts **for total 7%** of the records.
 - Additionally, when we perform imputation of each categorical feature independently, we **will risk losing important information** from other categorical features whose corresponding records may have useful data. It can also **lead to contribution of unknown bias and trends** in the dataset based on the result obtained after the imputation.
 - Thus, in our analysis, we **have not performed** any imputation of unknown values in categorical features.